

EXERCISE SHEET 15

GROUP AND RING THEORY

MAIN QUESTIONS - TO BE DONE BEFORE THE SEMINAR

15.1. Prove the following:

- (i) $\mathbb{Z}_6 \otimes_{\mathbb{Z}} \mathbb{Z}_7 = 0$;
- (ii) $\mathbb{Z}_m \otimes_{\mathbb{Z}} \mathbb{Z}_n \cong \mathbb{Z}_d$, where m, n are positive integers and $(m, n) = d$;
- (iii) $\mathbb{Z}_p \otimes_{\mathbb{Z}} p\mathbb{Z} \cong \mathbb{Z}_p$, for a prime p .

15.2. Let A be a torsion abelian group. That is, for each $a \in A$ there exists a positive integer n (depending on a) such that $na = 0$. Show that $A \otimes_{\mathbb{Z}} \mathbb{Q} = 0$.

15.3. Is $\mathbb{Z}/p\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}$ a subgroup of $\mathbb{Z}/p\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Q}$?

EXTRA QUESTIONS

15.4. Let $h : M_{1 \times 2}(\mathbb{Z}) \times M_{2 \times 1}(\mathbb{Z}) \rightarrow M_{2 \times 2}(\mathbb{Z})$ be defined by $h(x, y) = yx$, where $M_{1 \times 2}(\mathbb{Z})$ and $M_{2 \times 1}(\mathbb{Z})$ are right and left $M_2(\mathbb{Z})$ -modules respectively. Show that h is biadditive but not balanced.

15.5. Show that the tensor product is unique up to isomorphism.